

About this guide

A quick reference to the installed audio system in the Hormel Theatre — what each piece of equipment and software is, how it fits into our signal chain. When in doubt, ask the Audio Supervisor or the Stephenson house A1 before changing anything system-level.

Hormel runs full Dante infrastructure at 48 kHz throughout, with fully redundant primary/secondary Dante networking. The house console is a Yamaha CL5; playback is QLab; recording and virtual soundcheck run in Nuendo; outboard plugin processing lives in VST Rack; and the Dante layer underneath it all is configured with Dante Controller.

System at a glance

Console	Yamaha CL5 (FOH) — 48 kHz, Dante-native
Stage I/O	2× Yamaha Rio3224-D stage boxes (Band and FOH positions)
System processing	Meyer Galileo 616 speaker processor
PA	L-Acoustics: 4× ARCS Wide mains, 5× 5XT front fills, 4× X8 delays, 2× SB18 subs; 4× LA4X + QSC CX-Q 8K8 amplification
Playback	QLab on dedicated Mac Studio (+1× M1 Mac Mini; a second Mac Mini is on hand but currently idle) via Dante Virtual Soundcard and Focusrite Scarlett 18i20
Recording	Nuendo record computer on Dante — virtual soundcheck capable
Wireless mics	Shure Axient Digital: 5× AD4Q assigned to Hormel (20 ch), ADX1M bodypack pool, 2× ADX2 handhelds
Comms	Clear-Com Arcadia (AES67, shared house-wide); FreeSpeak II — 3× IP transceivers in Hormel; HXii beltpack pool
Personal monitors	Allen & Heath ME-1 pool via ME-U hub
Word clock	Focusrite RedNet A8R is the Dante clock leader (shared with Stephenson) — do not change
Show control	QLab with MSC/LTC integration to Lighting via MIDI through the Focusrite Scarlett 18i20.

Confirm current configuration with the department before advancing changes.

Yamaha CL5 — FOH Console

The heart of the Hormel system. Everything on stage and in the racks arrives here over Dante.

What it is	72 mono + 8 stereo input channels, 24 mix buses + 8 matrix, Dante-native with redundant primary/secondary ports.
Where it lives	FOH mix position, Hormel.
Sample rate	48 kHz — the whole Hormel network runs 48 kHz. Never change the console or Dante sample rate mid-season.
How signal arrives	Stage inputs land in the two Rio3224-D boxes and travel over Dante to the console. The console-side patch (which Dante channel feeds which fader) is set in the console's Dante input patch; the network-side patch is made in Dante Controller.
Remote control	CL Editor / StageMix over the dedicated audio-control Wi-Fi at FOH and band positions.
Outputs	Console outputs feed the Galileo 616 for PA processing and output distribution.

GOTCHAS

Save the show file to a USB stick after every programming session — the console itself cannot store show files directly. If a stage input is dead, check three layers in order: analog into the Rio (gain/48V on the right slot), the Dante subscription (Dante Controller), and the console input patch — a wrong layer diagnosis wastes the most troubleshooting time. Note the identical CL5 in Hardes is a separate physical console with a separate network identity — don't confuse the two in Dante Controller. The Hardes CL5 is not on the same network, so it does not appear in Dante Controller at all.

QLab 4.7 — Playback & Show Control

All sound-effect and music playback, plus the show-control bridge to Lighting.

What it is	Figure 53's cue-based playback and show-control software for macOS.
Where it lives	Dedicated Apple Mac Studio (primary) + 1× M1 Mac Mini actually running QLab, at the FOH/control position. A second M1 Mac Mini is on hand but currently unused for QLab.
How it reaches the console	Audio leaves QLab through Dante Virtual Soundcard (DVS) and the Focusrite Scarlett 18i20, arriving at the CL5 as "QLab DVS" Dante channels.
Show control	QLab converts incoming triggers to MSC for Lighting via MIDI through the Focusrite Scarlett 18i20 — nothing in this chain runs over OSC. The hybrid LTC/MSC chain proven on Dear Evan Hansen is the house-documented signal path.
Routing setup	The QLab-to-Dante routing lives in two editors: Audio Output Patch Editor and Audio Map Editor. (The Mic Cues editor isn't relevant here — mic cues aren't used at Hormel.)
Sync troubleshooting	Workspace Status Window and Timecode Status Window are the live diagnostic views for sync issues.

GOTCHAS

DVS depends on the Mac's network interface settings — if QLab audio vanishes from the console, verify DVS is running, locked to the network at 48 kHz, and that its Dante subscriptions are intact before touching QLab itself. Keep macOS and QLab updates frozen during a run. Only one Mac Mini currently runs QLab — worth setting up the second as a backup machine eventually, especially since most designers don't use VST.

VST Rack — Outboard Plugin Processing

The house's "virtual outboard rack" — plugin FX inserted into the console over Dante.

What it is	Yamaha/Steinberg's plugin host: runs VST3 plugins on a computer and presents them to the console as Dante insert points — reverbs, dynamics, and mastering-style processing without hardware outboard.
Where it lives	The QLab/VST computers on the audio-control network.
How it's used	Channels or buses on the CL5 are inserted out to VST Rack over Dante and returned the same way.
Session files	Save the VST Rack session alongside the console show file — the console insert patch and the VST Rack session must match or inserted channels go silent.

GOTCHAS

An inserted channel passes through the computer — if VST Rack isn't running (or the machine sleeps), that channel is silent, not dry. Disable sleep/App Nap on the host machine and verify every insert passes audio during preshow line check. The audio I/O unit next to the computers has a physical bypass switch — it must be switched off (not lit red) for inserts to pass through VST Rack correctly.

Nuendo — Recording & Virtual Soundcheck

Multitrack capture of every stage input, and playback of those recordings back into the console.

What it is	Steinberg's post-production DAW, used here as the multitrack record/playback machine.
Where it lives	Dedicated Nuendo record computer on the Dante network (via Yamaha Dante I/O).
Recording	Subscribes to the same Dante channels feeding the console, so every stage input can be captured during a run-through without touching the FOH mix.
Virtual soundcheck	Recorded tracks are played back into the console's input channels over Dante — mix, EQ, and program scenes with the real cast inputs, no cast on stage. Flip the console input patch (or Dante subscriptions) from stage sources to Nuendo returns, and flip it back when done.

GOTCHAS

The single most common virtual-soundcheck error is forgetting to restore the input patch to the live stage sources afterward — always verify the patch is back on Rio inputs before anyone walks on stage. Arm and check record levels before every capture. Watch disk space — a full show at 48 kHz multitrack is large.

Dante Controller — The Network Layer

The patchbay for everything above: routing, naming, clocking, and troubleshooting the Dante network.

What it is	Audinate's free configuration/monitoring app for Dante networks. Every device in this guide — CL5, Rios, DVS/QLab, Nuendo, AD4Q receivers, Arcadia — appears in its Routing view.
Core action	"Subscribing" a receive channel to a transmit channel is the network-side patch. The Routing/Network view shows every subscription at once.
Clocking	The Focusrite RedNet A8R is the clock leader for both Hormel and Stephenson. Hormel is 48 kHz throughout. Watch the Clock Status Monitor — an unstable clock = dropouts network-wide, not per-device.
Network design	Fully redundant primary/secondary Dante on dedicated audio VLANs (Dante traffic and audio control are separate VLANs). Addressing and credentials are deliberately not published in this guide — ask the Audio Supervisor.
Device Lock	Some devices (Arcadia, AD4Q) may carry a Dante Device Lock PIN — Dante Controller handles lock/unlock and PIN recovery.
First stops when broken	Device missing or audio not flowing: check IP/interface configuration first, then switch and cabling.

GOTCHAS

Dante Controller changes take effect immediately on the live network — there is no offline/undo. Don't rename devices or channels mid-season; names are how every saved patch finds its targets. Never change the preferred clock leader or a device's sample rate without the Audio Supervisor. Run Dante Controller only from a machine on the correct interface/VLAN — a laptop on the wrong NIC sees a half-empty network and invites wrong conclusions.

Where to Find More

If it isn't covered here, ask before guessing.

Venue rider	Hormel Audio System Rider — the outward-facing system spec for visiting designers/engineers. Current copy from the Audio Supervisor.
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Addressing & credentials	Deliberately not published in any document — ask the Audio Supervisor.
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Contact List

Key staff for the Hormel and the wider department. Blank cells to be filled in — pencil in numbers or send them to the Audio Supervisor to update this document.

Name	Role	Phone	Email
Matthew Gibney	Audio Supervisor		m.gibney@phoenixtheatre.com
<i>vacant</i>	A1 — Hormel		
MJ	Audio Apprentice		
Jess Baker	A1 — Stephenson		j.baker@phoenixtheatre.com
Karla Frederick	Production Manager		k.fredrick@phoenixtheatre.com

A1 — Hormel is vacant: Katie Paris departed after Dear Evan Hansen closed. Update this row once a replacement is named.

Department — email	sounddept@phoenixtheatre.com · ptaudiodept@gmail.com
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Theatre	The Phoenix Theatre Company — 1825 N Central Ave, Phoenix, AZ 85004
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Sound Designers are per-show external contractors (recent: Ashton Michael Corey — Les Mis; Dan Perelstein Jaquette — Dear Evan Hansen; Dave Temby — Ever After) — contact via the Audio Supervisor.